

Toward LLM-based Conversational Agents That Predict Well-being From Multimodal Conversations

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Traditionally, mental health and psychological well-being assessments depend on self-report questionnaires (e.g., PHQ-9), which are administered infrequently and may suffer from recall bias and errors. As mental health needs increasingly outpace clinical capacity, healthcare systems require low-burden, scalable approaches for early screening and monitoring. This study investigates the potential of conversational agents, using large language models (LLMs), to infer psychological well-being from naturalistic, multimodal behavior by three studies evaluating the performance of input modalities (text, acoustic, and facial cues), model architectures, and LLM prompting strategies. Our results show that LLMs with multimodal inputs, using lightweight prompting without task-specific fine-tuning, consistently outperform regression baselines, highlighting their potential to complement existing mental health services through more frequent, ecologically valid assessment with reduced patient burden and clinical workload.

CCS Concepts: • **Human-centered computing** → **User models**; • **Applied computing** → *Health care information systems*; • **Computing methodologies** → *Natural language generation*.

Additional Key Words and Phrases: Conversational Agent, LLMs, Mental Health, Well-being Assessment, Multimodal Analysis, Prompt Engineering, Affective Computing

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1 Introduction

Mental disorders remain a leading cause of global disability, imposing substantial costs on individuals, healthcare systems, and society at large [69]. Standard mental health and psychological assessments rely heavily on self-report questionnaires, such as PHQ-9 [28] and GAD-7 [54], which provide cost-effective and standardized guidelines for diagnosis and treatment planning. However, these instruments are often limited by retrospective recall bias and are poorly suited to capture the variability of mental health symptoms and the contextual factors that shape everyday well-being [37]. These limitations underscore the need for low-burden, scalable methods for assessing psychological well-being, a challenge that is both a pressing public health imperative and a fundamental scientific problem [40, 50]. One prominent response to these challenges has been the use of ecological momentary assessment (EMA) [51], which involves repeatedly sampling individuals' psychological states in their natural environments to capture in-situ experiences. However, EMA remains reliant on repetitive self-report and can introduce burden and disengagement over time [39]. These limitations have motivated interest in complementary approaches to infer mental health states from naturally generated text and multimodal behavioral data using AI and machine learning techniques. However, traditional regression-based models and neural network approaches often struggle with the nonlinear, high-dimensional nature of psychological signals and perform poorly on unstructured inputs, such as spontaneous speech or open-ended responses [9, 48, 56]. Recently, Large Language Models (LLMs) have shown promise for text-based mental health prediction, primarily targeting acute clinical risk, such as depression or suicide ideation, through task-specific fine-tuning or specialized prompting [21, 53, 60, 70]. Yet, their potential for broader well-being assessment from naturalistic, multimodal behavior remains underexplored [14, 17, 23].

We envision that an *LLM-driven conversational agent* that engages users through brief, natural, open-ended dialogue could offer a more natural and engaging way to monitor and assess people's mental health and well-being on a daily basis while complementing validated clinical instruments based on self-report questionnaires. As a critical first step, we investigated whether LLMs can be used for low-burden, conversational well-being assessment, and conducted three studies to evaluate how LLM-based approaches can be used to predict standardized well-being measures from multimodal signals extracted from self-recorded videos of open-ended responses. Specifically, we examined the following research questions:

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- **RQ1.** Does integrating visual and acoustic non-verbal cues improve LLM-based well-being assessment in comparison to transcript-only input?
- **RQ2.** How do LLM-based approaches compare to other machine learning techniques previously used in psychological assessment?
- **RQ3.** Do advanced prompting strategies further improve LLMs' performance beyond zero-shot inference?

2 Related Works

Digital Mental Health Assessment. Mental health research increasingly recognizes that well-being is dynamic and context-dependent [63], motivating a shift away from static, retrospective self-reports toward ecologically valid approaches that capture moment-to-moment fluctuations in everyday life [57, 63, 67]. Ecological Momentary Assessment [51] (EMA) enables such in-situ measurement and has shown to be valuable for studying dynamic constructs, such as stress and well-being, that are poorly captured by infrequent surveys [1, 35, 67]. EMAs are often used in conjunction with Just-in-Time Adaptive Interventions [36] (JITAI), which deliver timely and context-aware mental health intervention and support [1, 13, 20, 61, 63]. However, the effectiveness of these approaches is often constrained by declining user engagement, driven by repetitive prompts, survey fatigue, and limited expressiveness of structured self-reports [39]. These limitations highlight the need for monitoring approaches that can sustain engagement while reducing response burden. Thus, we propose that a conversational agent that infers people's well-being through naturalistic, open-ended conversations could be a promising alternative.

AI/ML-based Mental Health Prediction Mental states are inherently complex and expressed across multiple modalities [27, 42, 66]. Thus, Artificial Intelligence and machine learning (AI/ML) approaches to mental health have evolved from unimodal analyses of isolated signals, such as acoustic markers of depression, lexical indicators of distress, or reduced facial expressivity, to multimodal frameworks that integrate linguistic, acoustic, and visual cues for more robust inference [7, 42, 46, 56]. Traditional multimodal systems have largely relied on regression-based or shallow learning models, which often struggle with the nonlinear, high-dimensional nature of psychological data and unstructured inputs such as spontaneous speech [9, 48]. Large Language Models (LLMs) have recently shown strong performance in mental health prediction from text [24] and, more recently, multimodal inputs [23]. Although current work primarily targets acute clinical pathologies, such as depression and suicidal ideation [60, 70], and focuses on using clinical interviews [5, 73], comparatively little attention has been given to broader well-being assessment from naturalistic, multimodal behavior. To address this gap, our work evaluates untuned LLMs for multimodal well-being assessment from open-ended video responses. By systematically examining modality integration, model architecture, and prompting strategies, we demonstrate the potential of LLM-based conversational agents as scalable, low-burden tools for high-frequency, ecologically valid mental health monitoring beyond acute clinical risk. Beyond multimodal feature integration, LLM performance in both general language tasks [4] and psychometric assessment depends strongly on prompting strategy [52, 71]. Chain-of-Thought (CoT) prompting elicits intermediate reasoning steps [26, 68], which is critical for interpretability in mental health contexts [71]. Retrieval-Augmented Generation (RAG) addresses this limitation by grounding predictions in relevant reference cases or clinical benchmarks, improving calibration and reducing hallucinations [30, 52]. Thus, we evaluate each strategy separately, as well as their combination. Motivated by iterative thought-and-retrieval frameworks [59, 65], combining CoT with RAG integrates structured reasoning with empirical grounding, which may yield more reliable predictions across diverse and skewed well-being measures and better support longitudinal monitoring.

3 Data Collection for Well-being Prediction Task: Self-report Survey and Video Responses

We recruited $N=81$ participants (53 M, 26 F, 2 Other/Prefer not to say; age $M=19.95$, $SD=3.78$, $range=18-51$) from the [Anonymous] University's psychology department subject pool. Participants were racially diverse (50.6% White/Caucasian, 29.6% Asian, 6.2% Black/African-American, and 13.6% Multiracial, Hispanic, or Other), were mostly native English speakers (88.9%), and were enrolled in various undergraduate programs, including STEM and health sciences. The study was conducted entirely online via Qualtrics (qualtrics.com). Once the consent was electronically signed, participants completed nine validated self-report assessments regarding their overall mental health and well-being: Generalized Anxiety Disorder-7 (GAD-7) [54], Meaning in Life Questionnaire (MLQ) [55], Multidimensional Scale of Perceived Social Support (MSPSS) [74], Patient Health Questionnaire-9 (PHQ-9) [28], Perceived Stress Scale (PSS) [6], Scale of Positive and Negative Experience (SPANE) [12], Sleep Quality Scale (SQS) [72], Satisfaction With Life Scale (SWLS) [11], and the UCLA Loneliness Scale (UCLA) [47]. These self-report responses served as ground-truth labels for the well-being prediction task for our study.

Participants were asked to record themselves responding to seven open-ended questions regarding various aspects of their well-being (e.g., physical health, sleep, social support, purpose). Questions for video responses were designed to be

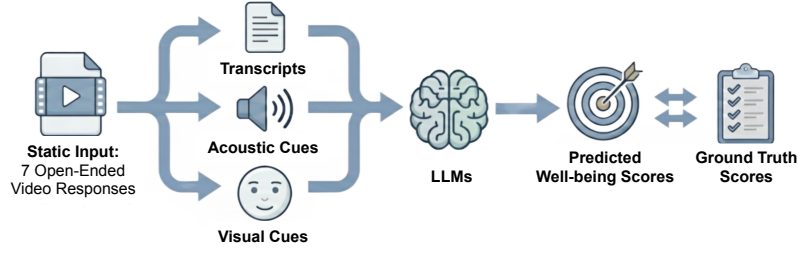


Fig. 1. Offline multimodal assessment pipeline. Each participant provided seven open-ended video responses, from which transcripts, acoustic cues, and visual cues are extracted and summarized as input to LLMs for predicting validated well-being survey scores.

open-ended and suitable for conversational tasks (e.g., “Please describe your health in general. Do you have any specific areas of concern?”). Our study was approved by the [anonymous] University Institutional Review Board, and each participant was compensated with either 1.5 credits in a course or a \$10 Amazon gift card, contingent upon passing internal attention and screening checks. Additional details on recruitment and participant demographics are provided in Section A. The full set of video-response prompts is listed in Section D.2, and the dataset partitioning used for training, validation, and testing across all evaluated methods and baselines is described in Section B.

4 Multimodal Feature Extraction

Since current LLMs operate primarily over textual inputs, we convert nonverbal video signals into compact, human-readable summaries that can be appended to the transcript in the prompt. For each participant, all approaches therefore use the same information aggregated across seven open-ended video responses, processed into three signal streams (Figure 1): **(1) transcripts**, generated with OpenAI’s Whisper speech recognition engine [43]; **(2) visual cues**, consisting of Facial Action Units (AUs) from the Facial Action Coding System [18], such as the frequency and intensity of smiles, brow raises, extracted using Affdex SDK [34]; and **(3) acoustic cues**, extracted with OpenSMILE [16], consisting of paraverbal features (e.g., pitch, intensity, jitter, speaking rate, pauses). Visual and acoustic cues are summarized using standard descriptive statistics (count, mean, standard deviation, min/max, quartiles) and combined with transcripts to form a unified, text-compatible multimodal input for LLM prompting. This aggregation strategy follows established psychometric evidence that stable psychological traits emerge reliably when behavior is aggregated across multiple situations rather than inferred from a single response [15].

5 Evaluation Metrics

We evaluate model performance against ground-truth survey scores using three complementary metrics. **Category accuracy** measures how accurately models classify participants into standard clinical severity bins derived from predicted scores. We additionally compute **Pearson correlation** (ρ) [41] to assess linear association between predicted and ground-truth continuous scores, and **Matthews Correlation Coefficient (MCC)** [33] as a balanced classification metric robust to class imbalance [10]. In the three main study sections (Sections 6 to 8), we report category accuracy results. MCC results are discussed in Section 9, and Pearson correlation results are provided Section G. For category accuracy, we include a **RANDGUESS** baseline, defined as the reciprocal of the number of severity classes ($1/C$) for each survey. **RANDGUESS** is excluded from Pearson correlation and MCC analyses, as random predictions yield no meaningful correlation-based or discriminative signal.

6 Study 1: Effects of Multimodal Cues on LLM-Based Well-Being Assessment

In this first study, we address **RQ1**: “Does integrating visual and acoustic non-verbal cues improve LLM-based well-being assessment in comparison to transcript-only input?”. In order to investigate this, we use a single, fixed large language model (LLaMA-4-Maverick [58]) with a fixed zero-shot prompt (provided in Section F) and systematically vary the input modality configuration from unimodal to multimodal while holding the model, using the same prompt structure, and participant data. This controlled design isolates the contribution of multimodal information to LLM-based inference of well-being.

Method. We compare a unimodal, transcript-only input against multimodal variants that combining the same transcripts with summarized facial cues, summarized acoustic cues, or both. Prior work [70] shows that transcript-only language features can predict mental health, motivating our transcript-only baseline. The unimodal setting (LLAMAT0) uses *only* transcripts from seven videos. The multimodal variants add *only* visual cues (LLAMATF0), *only* acoustic cues (LLAMATA0), or *both* (LLAMATFA0).

Table 1. **Category prediction accuracy** (mean $\pm$ std over three seeds) across input configurations. Best values per column are in **bold**, showing that full multimodal input (LLAMATFA0) show stronger gain for distress-related constructs (e.g., GAD-7, PHQ-9) than others.

Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
RANDGUESS	0.25	0.20	0.33	0.20	0.33	0.20	0.20	0.14	0.25
LLAMAT0	0.47 $\pm$ 0.00	0.50 $\pm$ 0.00	0.67 $\pm$ 0.04	0.51 $\pm$ 0.03	0.63 $\pm$ 0.03	0.40 $\pm$ 0.03	0.18 $\pm$ 0.00	0.33 $\pm$ 0.01	0.44 $\pm$ 0.00
LLAMATF0	0.47 $\pm$ 0.03	0.41 $\pm$ 0.05	0.75 $\pm$ 0.04	0.52 $\pm$ 0.02	0.77 $\pm$ 0.01	0.40 $\pm$ 0.03	0.17 $\pm$ 0.02	0.30 $\pm$ 0.01	0.47 $\pm$ 0.04
LLAMATA0	0.37 $\pm$ 0.02	0.46 $\pm$ 0.03	0.76 $\pm$ 0.01	0.40 $\pm$ 0.01	0.73 $\pm$ 0.02	0.41 $\pm$ 0.03	0.17 $\pm$ 0.03	0.33 $\pm$ 0.01	0.48 $\pm$ 0.01
LLAMATFA0	0.54 $\pm$ 0.04	0.50 $\pm$ 0.03	0.77 $\pm$ 0.01	0.54 $\pm$ 0.01	0.79 $\pm$ 0.01	0.40 $\pm$ 0.04	0.18 $\pm$ 0.01	0.32 $\pm$ 0.01	0.52 $\pm$ 0.04

Table 2. **Category prediction accuracy** (mean $\pm$ std over three seeds) across all models in three studies. Best values are in **bold**. LLM-based methods, especially with RAG+CoT prompting, show the most robust performance under class imbalance across well-being surveys.

	Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
Regres.	RANDGUESS	0.25	0.20	0.33	0.20	0.33	0.20	0.20	0.14	0.25
	RIDGE	0.31±0.00	0.54±0.00	0.56±0.00	0.25±0.00	0.41±0.00	0.31±0.00	0.72±0.00	0.06±0.00	0.44±0.00
	RANDFOREST	0.44±0.06	0.51±0.01	0.62±0.00	0.17±0.02	0.69±0.03	0.35±0.02	0.75±0.00	0.17±0.02	0.28±0.08
VLMs	LLAVA	0.09±0.01	0.31±0.00	0.26±0.01	0.20±0.04	0.82±0.13	0.42±0.01	0.02±0.01	0.18±0.01	0.35±0.06
	QWEN	0.42±0.03	0.63±0.00	0.72±0.03	0.53±0.03	0.74±0.01	0.43±0.01	0.10±0.01	0.35±0.01	0.52±0.04
LLMs	DEEPSEEK	0.52±0.04	0.38±0.01	0.75±0.03	0.53±0.04	0.70±0.04	0.42±0.04	0.14±0.01	0.27±0.04	0.53±0.05
	GPT	0.48±0.04	0.47±0.05	0.74±0.01	0.43±0.04	0.69±0.01	0.42±0.04	0.21±0.08	0.33±0.06	0.49±0.01
	LLAMATFA0	0.54±0.04	0.50±0.03	0.77±0.01	0.54±0.01	0.79±0.01	0.40±0.04	0.18±0.01	0.32±0.01	0.52±0.04

Results. Table 1 shows that LLM-based approaches consistently outperform RANDGUESS, indicating that naturalistic video responses contain meaningful psychometric signals that LLMs can interpret effectively. Transcript-only input (LLAMAT0) is sufficient for positive functioning constructs that are explicitly verbalized (e.g., MLQ, SWLS), whereas full multimodal integration (LLAMATFA0) performs best on distress-oriented scales (GAD-7, PHQ-9, PSS, UCLA), highlighting the greater importance of nonverbal cues in detecting psychological distress compared to other well-being domains. Isolated visual or acoustic inputs (LLAMATF0, LLAMATA0) provide only selective gains (e.g., MSPSS, PSS). In contrast, SQS remains challenging across all configurations, suggesting limitations in capturing sleep-related constructs in the current setting.

7 Study 2: Model Architecture Comparison

Results from Study 1 (Section 6) showed that incorporating multimodal inputs improves LLM-based well-being assessment. Thus, Study 2 aims to address **RQ2**: “How do LLM-based approaches compare to other machine learning techniques previously used in psychological assessment?”. To answer this question, we benchmark the best-performing LLM with multimodal feature representation from Study 1 against widely used alternative architectures, including supervised regression models and vision-language models (VLMs), under a consistent multimodal input configuration. This design allows us to isolate differences arising from model architecture and inference capability rather than input content or data availability.

Method. We benchmark LLAMATFA0 against two regression baselines, two VLMs, and two alternative LLMs. The **regression baselines**, Ridge [22] (RIDGE) and Random Forest [3] (RANDFOREST), use LIWC-22 [2] transcript features combined with the same acoustic and facial features as LLAMATFA0 (but without standard summary) and require task-specific training. We also evaluate two **VLMs**, Qwen2.5-VL [62] (QWEN) and LLaVA1.6:7B [31] (LLAVA), which process raw video frames (rather than summarized facial and acoustic cues) together with transcripts. Finally, we evaluate two additional **LLMs**, DeepSeek-R1:70B [8] (DEEPSEEK) and GPT-4o [38] (GPT). All LLM- and VLM-based models use the same zero-shot prompt (Section F).

Results. Table 2 shows that LLMs substantially outperform regression models and VLMs across nine well-being surveys, indicating that LLMs using transcripts and summarized non-verbal cues currently provide more stable and effective prediction. In contrast, regression models exhibit limited generalization: while they perform competitively on more structured or low-variance measures such as SQS, they struggle on constructs that depend on nuanced, context-dependent language and affective expression (e.g., GAD-7, PHQ-9). Although LLAVA achieves the highest accuracy on PSS, VLM-based methods underperform and show higher variance on most other surveys. Among LLMs, LLAMATFA0 performs best on most of the surveys, while DEEPSEEK and GPT remain competitive.

Table 3. **Category prediction accuracy** (mean $\pm$ std over three seeds) across LLM prompting strategies. Best values per column are in **bold**, demonstrating that RAG+CoT consistently outperforms other prompting methods across all nine well-being surveys.

Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
RANDGUESS	0.25	0.20	0.33	0.20	0.33	0.20	0.20	0.14	0.25
LLAMATFA0	0.54 $\pm$ 0.04	0.50 $\pm$ 0.03	0.77 $\pm$ 0.01	0.54 $\pm$ 0.01	0.79 $\pm$ 0.01	0.40 $\pm$ 0.04	0.18 $\pm$ 0.01	0.32 $\pm$ 0.01	0.52 $\pm$ 0.04
LLAMACoT	0.56 $\pm$ 0.03	0.66 $\pm$ 0.03	0.75 $\pm$ 0.03	0.55 $\pm$ 0.01	0.77 $\pm$ 0.03	0.44 $\pm$ 0.03	0.19 $\pm$ 0.03	0.34 $\pm$ 0.01	0.55 $\pm$ 0.05
LLAMAFEWSHOT	0.50 $\pm$ 0.01	0.66 $\pm$ 0.03	0.65 $\pm$ 0.01	0.47 $\pm$ 0.01	0.80 $\pm$ 0.01	0.42 $\pm$ 0.01	0.16 $\pm$ 0.04	0.27 $\pm$ 0.04	0.47 $\pm$ 0.01
LLAMARAG	0.55 $\pm$ 0.01	0.64 $\pm$ 0.03	0.66 $\pm$ 0.01	0.43 $\pm$ 0.01	0.75 $\pm$ 0.01	0.42 $\pm$ 0.01	0.56 $\pm$ 0.01	0.29 $\pm$ 0.01	0.49 $\pm$ 0.01
LLAMARAGCoT	0.57 $\pm$ 0.03	0.67 $\pm$ 0.01	0.77 $\pm$ 0.01	0.56 $\pm$ 0.03	0.82 $\pm$ 0.01	0.47 $\pm$ 0.04	0.56 $\pm$ 0.01	0.37 $\pm$ 0.01	0.58 $\pm$ 0.01

Table 4. **Matthews Correlation Coefficient (MCC)** (mean $\pm$ std over three seeds) across all models in three studies. Best values per column are in **bold**, showing LLM-based methods, especially when using RAG+CoT prompting technique, achieve higher and more consistent MCC than regression (Regres.) and VLM approaches across most well-being surveys, indicating better robustness under class imbalance.

Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
Regres.									
RIDGE	0.16 $\pm$ 0.01	0.06 $\pm$ 0.01	0.20 $\pm$ 0.01	-0.03 $\pm$ 0.00	0.06 $\pm$ 0.01	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01	0.08 $\pm$ 0.01
RANDFOREST	0.41 $\pm$ 0.01	0.09 $\pm$ 0.01	0.07 $\pm$ 0.01	0.03 $\pm$ 0.00	0.37 $\pm$ 0.01	-0.08 $\pm$ 0.02	0.21 $\pm$ 0.02	-0.04 $\pm$ 0.00	0.06 $\pm$ 0.01
VLMs									
LLaVA	0.06 $\pm$ 0.00	0.04 $\pm$ 0.00	0.18 $\pm$ 0.02	0.13 $\pm$ 0.13	-0.03 $\pm$ 0.04	0.17 $\pm$ 0.02	-0.05 $\pm$ 0.04	-0.02 $\pm$ 0.02	0.08 $\pm$ 0.10
QWEN	0.12 $\pm$ 0.05	0.00 $\pm$ 0.00	0.42 $\pm$ 0.05	0.31 $\pm$ 0.04	0.34 $\pm$ 0.05	0.15 $\pm$ 0.03	-0.01 $\pm$ 0.02	0.19 $\pm$ 0.02	0.36 $\pm$ 0.04
LLMs									
LLAMAT0	0.19 $\pm$ 0.00	0.21 $\pm$ 0.00	0.34 $\pm$ 0.07	0.30 $\pm$ 0.04	0.33 $\pm$ 0.06	0.15 $\pm$ 0.05	-0.04 $\pm$ 0.00	0.17 $\pm$ 0.02	0.28 $\pm$ 0.01
LLAMATF0	0.12 $\pm$ 0.06	-0.02 $\pm$ 0.09	0.46 $\pm$ 0.11	0.32 $\pm$ 0.04	0.41 $\pm$ 0.02	0.14 $\pm$ 0.05	-0.04 $\pm$ 0.04	0.17 $\pm$ 0.02	0.28 $\pm$ 0.06
LLAMATA0	0.15 $\pm$ 0.02	0.10 $\pm$ 0.08	0.48 $\pm$ 0.03	0.21 $\pm$ 0.03	0.30 $\pm$ 0.05	0.16 $\pm$ 0.05	-0.01 $\pm$ 0.03	0.20 $\pm$ 0.02	0.34 $\pm$ 0.03
DEEPSEEK	0.20 $\pm$ 0.01	0.01 $\pm$ 0.05	0.46 $\pm$ 0.06	0.31 $\pm$ 0.08	0.41 $\pm$ 0.07	0.18 $\pm$ 0.02	0.04 $\pm$ 0.01	0.11 $\pm$ 0.04	0.32 $\pm$ 0.08
GPT	0.24 $\pm$ 0.06	0.14 $\pm$ 0.10	0.47 $\pm$ 0.03	0.25 $\pm$ 0.05	0.35 $\pm$ 0.03	0.21 $\pm$ 0.03	0.06 $\pm$ 0.07	0.19 $\pm$ 0.09	0.36 $\pm$ 0.02
LLAMATFA0	0.20 $\pm$ 0.03	0.16 $\pm$ 0.07	0.55 $\pm$ 0.02	0.29 $\pm$ 0.03	0.52 $\pm$ 0.04	0.14 $\pm$ 0.05	-0.01 $\pm$ 0.02	0.18 $\pm$ 0.02	0.29 $\pm$ 0.07
LLAMACoT	0.21 $\pm$ 0.02	0.29 $\pm$ 0.05	0.49 $\pm$ 0.06	0.34 $\pm$ 0.03	0.39 $\pm$ 0.07	0.20 $\pm$ 0.05	0.01 $\pm$ 0.01	0.18 $\pm$ 0.00	0.41 $\pm$ 0.07
LLAMAFEWSHOT	0.25 $\pm$ 0.07	0.19 $\pm$ 0.08	0.15 $\pm$ 0.11	0.28 $\pm$ 0.03	0.43 $\pm$ 0.06	0.14 $\pm$ 0.03	0.04 $\pm$ 0.08	0.13 $\pm$ 0.05	0.16 $\pm$ 0.01
LLAMARAG	0.13 $\pm$ 0.03	0.12 $\pm$ 0.10	0.20 $\pm$ 0.00	0.12 $\pm$ 0.04	0.16 $\pm$ 0.00	0.14 $\pm$ 0.03	-0.02 $\pm$ 0.00	0.13 $\pm$ 0.03	0.30 $\pm$ 0.02
LLAMARAGCoT	0.19 $\pm$ 0.05	0.29 $\pm$ 0.03	0.54 $\pm$ 0.04	0.36 $\pm$ 0.04	0.51 $\pm$ 0.06	0.25 $\pm$ 0.07	0.14 $\pm$ 0.14	0.27 $\pm$ 0.09	0.40 $\pm$ 0.00

8 Study 3: LLM Prompting Strategies Comparison

Having established that LLMs with multimodal feature representation outperform other model architectures, and given prior works that advanced prompting can enhance LLM reasoning in mental health area [29, 52, 71], we next address **RQ3**: “Do advanced prompting strategies further improve LLMs’ performance beyond zero-shot inference?”. To answer this question, we compare multiple prompting strategies, from direct zero-shot to retrieval-augmented and reasoning-based approaches, using the best-performing multimodal input from Study 1 (Section 6) and a fixed LLaMA-4-Maverick model [58], isolating the effect of prompting on our analysis.

Method. The most direct prompting technique, **zero-shot** [4], previously used for all LLM and VLM baselines in Sections 6 and 7, includes a brief survey context and a short rationale in the prompt. We then evaluate whether more advanced prompting techniques can improve performance by comparing four prompting strategies [32] against zero-shot prompting, LLAMATFA0, when using the same full multimodal input and LLaMA-4-Maverick. **Chain-of-Thought (CoT)** [68] elicits an explicit reasoning trace before prediction (LLAMACoT). **Few-shot** [4] adds five fixed input-score examples from the regression training set (LLAMAFEWSHOT). The selected examples differ by survey and are chosen to cover the full range of severity categories for each assessment, which are provided in Section F.3. More advanced, **retrieval-augmented (RAG)** prompting [30] dynamically retrieves the top- k similar examples ($k=5$) per participant, from the regression training set, using multimodal embeddings retrieved from inputs (LLAMARAG). **RAG+CoT** combines RAG and CoT strategies (LLAMARAGCoT). Details of the RAG implementation and example prompts for each prompting strategy are provided in Sections E and F, respectively.

Results. Table 3 shows that LLAMARAGCoT achieves the best or tied-best performance across all nine surveys. In contrast, LLAMAFEWSHOT and LLAMACoT yield inconsistent gains, likely due to mismatched fixed examples in few-shot prompting and ungrounded reasoning in CoT. Among example-based strategies, LLAMAFEWSHOT and LLAMARAG perform comparably on most surveys, except for SQS, where RAG raises accuracy significantly from the RANDGUESS level.

9 Discussion

As a preliminary investigation, this study is limited by its dataset size and a predominantly college-aged, non-clinical sample. This demographic focus resulted in a positively skewed distribution across several distress-oriented scales (Section C), which

may constrain the generalizability of these findings to broader clinical populations with more acute symptoms. To account for this imbalance, in addition to category accuracy, we report the Matthews Correlation Coefficient (MCC) [33], which is regarded as a robust measure of classification performance under class imbalance [10], across all models in three studies (Table 4).

LLMs can surpass random guessing level, indicating their ability to extract and interpret meaningful psychometric signals from naturalistic video responses. Modality importance is construct-dependent (Section 6): while transcripts alone may suffice for explicitly articulated positive functioning (e.g., SWLS, MLQ), distress-related measures (e.g., GAD-7, PHQ-9, PSS, UCLA) benefit more substantially from multimodal cues than other well-being surveys. These findings are supported by prior works linking depression to reduced facial expressiveness [42, 66] and characteristic acoustic patterns [7, 49]. MCC results reinforce this conclusion, highlighting the value of LLM-driven agents processing multimodal inputs for capturing psychological states that are not reliably conveyed through language alone.

Across model architectures, LLMs outperform regression-based models and VLMs (Section 7), which likely arises mostly from the ability to interpret open-ended response transcripts and jointly reason over multimodal cues in LLMs. Regression models exhibit domain-specific strengths on structured, low-variance measures such as SQS, where repetitive lexical cues or stable acoustic markers of fatigue are salient. However, they generalize poorly to constructs requiring nuanced, context-sensitive interpretation of language and affect (e.g., MSPSS, PHQ-9) [9, 48, 56], and their near-zero or negative MCC values indicate limited discriminative power. VLMs show similar limitations. For instance, although QWEN attains a seemingly competitive category accuracy of 63% on the Meaning in Life (MLQ) scale, its MCC is 0 because it fails to distinguish between severity level “Moderate” (12/32) and “High” (20/32) meaning ground-truth cases, effectively collapsing predictions into a single dominant class. This discrepancy highlights that high accuracy can mask poor discriminative power under class imbalance. The weak performance of VLMs likely reflects their lack of perception and reasoning [19]: while VLMs can capture visual context, they struggle to translate these perceptions into the multi-step, calibrated reasoning required for psychometric assessment. Overall, LLMs maintain higher and more stable MCCs, suggesting they capture underlying psychological signals rather than defaulting to population-level statistical baselines.

Prompting strategy further shapes the richness and reliability of extracted information in LLMs (Section 8). In particular, while CoT promotes structured inference, its benefits are limited without calibration, RAG provides this grounding, most notably for subtle signals from our structured multimodal inputs such as sleep quality (SQS), where accuracy improves from the random guessing level (20%) to over 50%. The consistently strong, best or tied-best, performance of RAG+CoT across both category accuracy and MCC demonstrates that combining structured reasoning with retrieval-based calibration yields more reliable and stable predictions across surveys. Evaluation using Pearson correlation (ρ) mirrors these similar patterns (Section G). Together, these results demonstrate that LLM-based inference, particularly when utilizing multimodal features and RAG+CoT prompting, is well-suited for longitudinal assessment, maintaining essential calibration as individual psychological states evolve over time.

Despite dataset limitations, our findings demonstrate the feasibility of multimodal, LLM-driven conversational agents as low-burden supporting tools for well-being assessment. Future work will extend to more diverse populations and evaluate interactive deployment, examining whether such agents can support more frequent and less time-consuming assessments, sustain user engagement, and ultimately reduce clinical workload. Rather than replacing gold-standard clinical assessments or EMA, these agents are best viewed as complementary, capturing rich contextual signals between clinical visits, especially when traditional EMA prompts are skipped or ignored. To ensure safe real-world deployment, these systems should incorporate reliability engineering principles, such as uncertainty-aware outputs and human-in-the-loop verification. Analogous to a smoke detector, these agents do not diagnose or treat conditions, but instead flag meaningful deviations in well-being that warrant follow-up with validated instruments or clinician review.

10 Conclusion

This work shows that LLM-based agents with multimodal feature representations can support low-burden well-being assessment across nine validated surveys. While transcripts alone capture explicitly articulated positive functioning, integrating nonverbal cues substantially improves sensitivity to psychological distress. Beyond input modality, we show that retrieve-then-reason (RAG+CoT) prompting yields the most robust and well-calibrated performance by grounding inference in relevant reference cases, particularly for subtle or weakly expressed constructs. Together, these results position conversational agents as a scalable, ecologically valid complement to existing monitoring frameworks, enabling high-frequency, low-burden assessment between clinical visits and supporting more proactive, personalized mental health care.

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A Recruitment and Dataset Quality

Recruitment and Ethical Oversight. Participant recruitment was conducted under the supervision of the Principal Investigator and approved by the [Anonymous] University’s Institutional Review Board. Undergraduate and graduate students were recruited via direct email invitations and the university psychology participant pool. Recruitment materials emphasized participation in mental health research and the opportunity to practice professional-style video interviews. Participation was fully voluntary, and participants could withdraw at any time without penalty. All participants provided informed consent prior to data collection and were compensated upon completion contingent on passing attention and screening checks.

Table 5. Participant demographic distribution ($N = 81$). Percentages may not sum to 100% due to rounding.

Characteristic	Value
Age (years)	$M = 19.95$, $SD = 3.78$ (range: 18–51)
Gender	
Male	53 (65.4%)
Female	26 (32.1%)
Other / Prefer not to say	2 (2.5%)
Race / Ethnicity	
White / Caucasian	41 (50.6%)
Asian	24 (29.6%)
Black / African American	5 (6.2%)
Multiracial / Hispanic / Other	11 (13.6%)
Native English Speaker	72 (88.9%)
Academic Status	
Undergraduate	Majority
Graduate	Minority

Data Quality and Retention. All retained participants completed the full study protocol, including nine validated self-report assessments and seven open-ended video responses. Participants who failed internal attention checks, provided incomplete survey data, or did not complete all required video recordings were excluded prior to analysis. The final dataset includes $N = 81$ participants, each contributing a complete multimodal record. A summary of participant demographics is provided in Table 5.

B Data Splits and Distribution

To support model development and unbiased evaluation, participants were partitioned into three disjoint sets for training, validation, and testing:

- **Training Set** ($N=40$): Participants P01-P38, P46, and P47. This set was used to train regression baselines and served as the reference pool for few-shot and RAG prompting.
- **Validation Set** ($N=9$): Participants P39-P45, P48, and P49. This set was used for the validation set when training regression baselines.
- **Test Set** ($N=32$): Participants P50-P81. This held-out set was used exclusively for final performance reporting across all models.

Participant identifiers were randomly assigned and do not encode demographic or behavioral information.

C Ground-Truth Score Distribution

Ground-truth scores were calculated based on the scoring guidelines of the standardized self-report instruments. As shown in Table 6, distress-related measures (e.g., GAD-7, PHQ-9) are skewed toward lower severity ranges, reflecting a predominantly non-clinical sample. In contrast, positive functioning measures (e.g., MLQ, SWLS) exhibit broader variance.

D Video Recording Requirements and Procedure

D.1 Video Recording Requirements

Immediately after providing informed consent and basic demographic information—and before completing any self-report surveys used as ground-truth labels—participants were required to complete a video recording check to ensure data quality and consistency across recordings. The following instructions were presented verbatim to all participants:

Table 6. Distribution of ground-truth well-being scores across the nine validated instruments. Scores reflect a predominantly non-clinical sample, with greater variance observed in positive functioning measures than in distress-oriented scales.

Survey	Scale Range	Mean	SD	Min-Max
GAD-7	0–21	4.16	4.25	0–17
PHQ-9	0–27	5.93	5.02	0–22
PSS	0–40	17.49	6.63	1–37
SQS	0–84	59.57	7.89	46–80
UCLA	20–80	42.06	10.12	24–70
MLQ	10–70	43.63	9.17	16–60
SPANE	–24–24	7.85	7.40	–11–24
SWLS	5–35	23.12	7.19	5–35
MSPSS	12–84	65.36	10.80	41–82

1. Have your face well lit (i.e. no harsh lighting or shadows) and please DO NOT wear a mask
2. Have your face in the center of the screen at all times
3. Make sure your audio can be clearly heard and there are no background noises
4. Please wear headphones to ensure audio quality
5. The video recording must be at least 60 seconds long

This recording check served to verify adequate lighting, facial visibility, and audio clarity prior to the main data collection phase.

D.2 Video Recording Procedure

Immediately before the first video-response prompt, participants were reminded with the following instruction:

In the following section, you will be answering 7 video questions. Each response should take at least 1 minute each.

Participants then recorded video responses to seven open-ended prompts, in order, designed to elicit naturalistic speech, paralinguistic cues, and facial behavior relevant to well-being assessment:

1. Please describe your health in general. Do you have any specific areas of concern?
2. How many hours do you sleep in a night on average? Do you have trouble sleeping? Please describe your sleep habits and if you are sleeping well through the night.
3. Do you prefer cooking at home or eating out? Do you feel like you are eating healthy? What does your ideal meal look like? Please describe more about what you eat on a typical basis.
4. Do you have people you could talk to about important events in your life? Are your relationships with friends and family supportive? Do you have supportive relationships? Please describe more about these relationships.
5. Do you wake up feeling excited about the day ahead? How satisfied are you with your life? Could you share more about how your life is going?
6. Do you feel like you have a sense of purpose and meaning in your life? Why or why not?
7. How have you been spending your free time lately? Do you often feel lonely?

Dataset Summary Statistics. The final dataset comprises 11.42 hours of video recordings. On average, each participant contributed approximately 1,250 words across all responses, with a mean response duration of 72.3 seconds per question and an average speech rate of 147.0 words per minute.

E RAG Implementation Details

To retrieve the most similar examples for few-shot learning, we employed a semantic similarity-based retrieval approach using dense vector embeddings. Each example in the training corpus was represented as a concatenated text string combining transcript, facial, and acoustic features. These text representations were encoded into 384-dimensional dense vector embeddings using the all-MiniLM-L6-v2 [64] SentenceTransformer model [44, 45]. All embeddings were L2-normalized to unit length to enable efficient cosine similarity computation. We constructed a searchable index using FAISS (Facebook AI Similarity Search) [25] with an IndexFlatIP (Inner Product) index, which performs exact nearest neighbor search over the normalized embedding space. Since the inner product on L2-normalized vectors is equivalent to cosine similarity, this configuration

enables efficient retrieval of semantically similar examples. During inference, the query participant’s full multimodal data block (comprising all seven question responses with their associated transcript, facial, and acoustic features) was embedded using the same encoder and searched against the pre-built index to retrieve the top-5 most similar examples based on cosine similarity. These retrieved examples were then incorporated into the few-shot prompt to provide contextual guidance for the model’s predictions.

F LLM Prompt Templates and Examples

This section outlines the five prompting strategies employed across all clinical survey evaluations. For each strategy, we provide the general format used for all surveys (e.g., PHQ-9, GAD-7) and use the MSPSS for illustration.

F.1 Zero-shot

Establishes a professional persona and enforces a strict output format to facilitate automated data extraction without intermediate explanation.

F.1.1 General Format.

You are a clinical psychologist evaluating a participant's well-being using the {SURVEY_NAME}.

SCALE OVERVIEW

Score Range: {RANGE}

Your task:

- 1) Analyze the participant's verbal responses, facial summary (AU), and speaking behavior.
- 2) Provide a score based on the evidence.
- 3) Provide interpretation and confidence.

IMPORTANT: Start directly with '{SURVEY_NAME} Score ({RANGE}):' (no intro text).

Output Format (4 lines, no markdown):

{SURVEY_NAME} Score ({RANGE}): [score]

Interpretation: [interpretation]

Confidence Level: [High / Moderate / Low]

Justification: [1-2 sentences citing verbal + nonverbal cues]

Now analyze the following participant:

{full_data}

F.1.2 Specific Implementation: MSPSS.

You are a clinical psychologist evaluating a participant's well-being using the **Multidimensional Scale of Perceived Social Stress**.

MSPSS SCALE OVERVIEW

The MSPSS is a validated assessment tool. Score Range: 12-84.

Your task:

1. Analyze the participant's verbal responses, facial expression summaries (AU), and speaking behavior.
2. Provide a MSPSS score based on the evidence.
3. Interpret the result and provide confidence level.

IMPORTANT: Start your response directly with 'MSPSS Score (12-84):'

Output Format:

MSPSS Score (12-84): [score]

Interpretation: [interpretation]

Confidence Level: [High/Moderate/Low]

Justification: [1-2 sentence explanation]

Now analyze the following participant:
{full_data}

F.2 Chain-of-Thought (CoT)

Directs the model to perform a step-by-step clinical breakdown, improving accuracy in complex multi-dimensional scales.

F.2.1 General Format.

You are a clinical psychologist evaluating a participant using the {SURVEY_NAME}.
Score Range: {RANGE}

ANALYSIS INSTRUCTIONS: Let's think step by step to ensure an accurate assessment.

STEP-BY-STEP REASONING

- 1) Read all responses; identify survey-relevant themes.
- 2) Use verbal + AU summary + speaking behavior.
- 3) Write item-level reasoning aligned with the construct.
- 4) Show total calculation logic.
- 5) Explain the final interpretation.

FINAL ANSWER (REQUIRED FORMAT; exactly 5 lines; no extra text):

CoT Reasoning: <item-level reasoning + total calculation + interpretation>
{SURVEY_NAME} Score: <integer within {RANGE}>
Interpretation: <Low / Moderate / High>
Confidence Level: <High / Moderate / Low>
Justification: <brief cue-based summary>

Now analyze the following participant:
{full_data}

F.2.2 Specific Implementation: MSPSS..

[Role and Overview same as MSPSS Zero-Shot]

You are a clinical psychologist evaluating a participant's well-being using the **Multidimensional Scale of Perceived Social Support**.

MSPSS SCALE OVERVIEW

The MSPSS is a validated assessment tool. Score Range: 12-84.

ANALYSIS INSTRUCTIONS: Let's think step by step.

1. Read all responses for social support themes.
2. Analyze verbal, AU summaries, and behavior.
3. Write Detailed Item Analysis for each MSPSS dimension.
4. Show total calculation process.

FINAL ANSWER (REQUIRED FORMAT)

CoT Reasoning: <detailed item analysis + calculation>
MSPSS Score: <12-84>
Interpretation: <Low/Moderate/High>
Confidence Level: <High/Moderate/Low>
Justification: <summary of indicators>

Now analyze the following participant:
{full_data}

Table 7. Few-shot example participants used for each survey. Fixed example sets differ across surveys and were chosen to represent the full distribution of severity categories for each scale.

Survey	Few-shot Examples
GAD-7	P04, P16, P30, P06, P03
PHQ-9	P04, P06, P15, P21, P46
PSS	P04, P06, P15, P46, P21
SQS	P15, P04, P33, P02, P01
UCLA	P04, P06, P13, P01, P02
MLQ	P18, P12, P13, P15, P01
SPANE	P21, P15, P30, P06, P01
SWLS	P21, P30, P06, P01, P04
MSPSS	P13, P06, P04, P02, P01

F.3 Few-shot

Incorporates static, five expert-labeled examples to calibrate the model’s scoring thresholds into a Zero-shot format. The selected examples vary across surveys and are intended to represent the complete spectrum of severity categories for each instrument. The specific few-shot example sets for all surveys are detailed in Table 7.

F.3.1 General Format.

You are a clinical psychologist evaluating a participant using the {SURVEY_NAME}.

Score Range: {RANGE}

Your task:

- 1) Use verbal + AU summary + speaking behavior.
- 2) Provide a score, interpretation, and confidence.

IMPORTANT: Start directly with '{SURVEY_NAME} Score ({RANGE}):' (no intro text).

Output Format (4 lines, no markdown):

{SURVEY_NAME} Score ({RANGE}): [score]

Interpretation: [interpretation]

Confidence Level: [High / Moderate / Low]

Justification: [1-2 sentences citing cues]

Few-shot examples for reference:

{examples}

Now analyze the following participant:

{full_data}

F.3.2 Specific Implementation: MSPSS..

[Role and Overview same as MSPSS Zero-Shot]

You are a clinical psychologist evaluating a participant's well-being using the **Multidimensional Scale of Perceived Social Support**.

MSPSS SCALE OVERVIEW

The MSPSS is a validated assessment tool. Score Range: 12-84.

ANALYSIS INSTRUCTIONS: Let's think step by step.

1. Read all responses for social support themes.
2. Analyze verbal, AU summaries, and behavior.
3. Write Detailed Item Analysis for each MSPSS dimension.
4. Show total calculation process.

Few-shot examples for reference:

{examples}

Now analyze the following participant:

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 {full_data}

F.4 Retrieval-Augmented Generation (RAG)

The prompting format is the same as Few-shot, but dynamically injects the most relevant historical cases from the dataset via vector similarity search.

F.4.1 General Format.

You are a clinical psychologist evaluating a participant using the {SURVEY_NAME}.
 Score Range: {RANGE}

Your task:

- 1) Use verbal + AU summary + speaking behavior.
- 2) Provide a score, interpretation, and confidence.

IMPORTANT: Start directly with '{SURVEY_NAME} Score ({RANGE}):' (no intro text).

Output Format (4 lines, no markdown):
 {SURVEY_NAME} Score ({RANGE}): [score]
 Interpretation: [interpretation]
 Confidence Level: [High / Moderate / Low]
 Justification: [1-2 sentences citing cues]

Retrieved Examples for reference (RAG):
 {retrieved_examples}

Now analyze the following participant:
 {full_data}

F.4.2 Specific Implementation: MSPSS..

[Role and Overview same as MSPSS Zero-Shot]

You are a clinical psychologist evaluating a participant's well-being using the
 Multidimensional Scale of Perceived Social Support (MSPSS).

MSPSS SCALE OVERVIEW

The MSPSS is a validated assessment tool. Score Range: 12-84.

ANALYSIS INSTRUCTIONS: Let's think step by step.

1. Read all responses for social support themes.
2. Analyze verbal, AU summaries, and behavior.
3. Write Detailed Item Analysis for each MSPSS dimension.
4. Show total calculation process.

 Retrieved Examples for reference (RAG):
 {retrieved_examples}

 Now analyze the following participant:
 {full_data}

F.5 RAG+CoT

Combines the logical rigor of CoT with the evidence-based grounding of retrieved cases.

F.5.1 General Format.

You are a clinical psychologist evaluating a participant using the {SURVEY_NAME}.

Score Range: {RANGE}

ANALYSIS INSTRUCTIONS: Let's think step by step to ensure an accurate assessment.

STEP-BY-STEP REASONING

- 1) Read all responses; identify survey-relevant themes.
- 2) Use verbal + AU summary + speaking behavior.
- 3) Write item-level reasoning aligned with the construct.
- 4) Show total calculation logic.
- 5) Explain the final interpretation.

FINAL ANSWER (REQUIRED FORMAT; exactly 5 lines; no extra text):

CoT Reasoning: <item-level reasoning + retrieved-case grounding + total + interpretation>

{SURVEY_NAME} Score: <integer within {RANGE}>

Interpretation: <Low / Moderate / High>

Confidence Level: <High / Moderate / Low>

Justification: <brief cue-based summary>

Retrieved Examples for reference (RAG):

{retrieved_examples}

Now analyze the following participant:

{full_data}

F.5.2 Specific Implementation: MSPSS..

[CoT MSPSS Template]

You are a clinical psychologist evaluating a participant's well-being using the **Multidimensional Scale of Perceived Social Support**.

MSPSS SCALE OVERVIEW

The MSPSS is a validated assessment tool. Score Range: 12-84.

ANALYSIS INSTRUCTIONS: Let's think step by step.

1. Read all responses for social support themes.
2. Analyze verbal, AU summaries, and behavior.
3. Write Detailed Item Analysis for each MSPSS dimension.
4. Show total calculation process.

FINAL ANSWER (REQUIRED FORMAT)

CoT Reasoning: <detailed item analysis + calculation>

MSPSS Score: <12-84>

Interpretation: <Low/Moderate/High>

Confidence Level: <High/Moderate/Low>

Justification: <summary of indicators>

Retrieved Examples for reference (RAG):

{retrieved_examples}

Now analyze the following participant:

{full_data}

G Additional Results

Overall, the Pearson correlation results in Table 8 align closely with the trends observed for category accuracy (Tables 1 to 3) and MCC (Table 4) in the main paper, demonstrating consistent performance across evaluation metrics. LLM-based methods show a stronger capacity to capture both continuous psychometric variation (Pearson's ρ) and categorical severity distinctions

Table 8. **Pearson Correlation (ρ)** (mean $\pm$ std over three seeds) across all models in thee studies. Best values per column are in **bold**, showing LLM-based methods, especially when using RAG+CoT prompting technique, achieve superior performance in capturing continuous psychometric signals than regression (Regres.) and VLM approaches across most well-being surveys

Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
Regres.									
RIDGE	0.25 $\pm$ 0.06	0.72 $\pm$ 0.24	-0.03 $\pm$ 0.00	0.06 $\pm$ 0.11	-0.20 $\pm$ 0.15	0.19 $\pm$ 0.00	0.20 $\pm$ 0.08	0.17 $\pm$ 0.04	-0.14 $\pm$ 0.00
RANDFOREST	0.43 $\pm$ 0.10	0.67 $\pm$ 0.33	-0.02 $\pm$ 0.07	0.24 $\pm$ 0.21	-0.22 $\pm$ 0.16	-0.13 $\pm$ 0.07	0.41 $\pm$ 0.12	-0.07 $\pm$ 0.21	-0.28 $\pm$ 0.07
VLMs									
LLAVA	0.14 $\pm$ 0.03	0.35 $\pm$ 0.01	-0.18 $\pm$ 0.02	-0.01 $\pm$ 0.36	0.04 $\pm$ 0.07	0.43 $\pm$ 0.16	0.24 $\pm$ 0.29	0.47 $\pm$ 0.08	0.17 $\pm$ 0.04
QWEN	0.50 $\pm$ 0.02	0.09 $\pm$ 0.11	0.51 $\pm$ 0.02	0.59 $\pm$ 0.04	0.57 $\pm$ 0.01	0.36 $\pm$ 0.07	0.31 $\pm$ 0.01	0.66 $\pm$ 0.02	0.36 $\pm$ 0.04
LLMs									
LLAMAT0	0.44 $\pm$ 0.02	-0.04 $\pm$ 0.01	0.51 $\pm$ 0.00	0.64 $\pm$ 0.01	0.66 $\pm$ 0.01	0.55 $\pm$ 0.01	0.37 $\pm$ 0.00	0.75 $\pm$ 0.01	0.63 $\pm$ 0.00
LLAMATF0	0.54 $\pm$ 0.04	0.13 $\pm$ 0.05	0.47 $\pm$ 0.05	0.62 $\pm$ 0.03	0.67 $\pm$ 0.02	0.46 $\pm$ 0.02	0.50 $\pm$ 0.04	0.67 $\pm$ 0.00	0.59 $\pm$ 0.01
LLAMATA0	0.47 $\pm$ 0.02	-0.14 $\pm$ 0.02	0.59 $\pm$ 0.01	0.59 $\pm$ 0.03	0.61 $\pm$ 0.02	0.57 $\pm$ 0.03	0.33 $\pm$ 0.00	0.71 $\pm$ 0.00	0.66 $\pm$ 0.02
DEEPSEEK	0.47 $\pm$ 0.07	-0.08 $\pm$ 0.19	0.55 $\pm$ 0.02	0.69 $\pm$ 0.05	0.64 $\pm$ 0.06	0.54 $\pm$ 0.01	0.28 $\pm$ 0.05	0.72 $\pm$ 0.02	0.62 $\pm$ 0.07
GPT	0.46 $\pm$ 0.02	0.14 $\pm$ 0.12	0.55 $\pm$ 0.03	0.67 $\pm$ 0.04	0.61 $\pm$ 0.06	0.53 $\pm$ 0.05	0.22 $\pm$ 0.04	0.72 $\pm$ 0.02	0.61 $\pm$ 0.02
LLAMATFA0	0.39 $\pm$ 0.11	0.05 $\pm$ 0.02	0.55 $\pm$ 0.02	0.61 $\pm$ 0.01	0.69 $\pm$ 0.03	0.51 $\pm$ 0.06	0.38 $\pm$ 0.01	0.73 $\pm$ 0.01	0.56 $\pm$ 0.11
LLAMACoT	0.47 $\pm$ 0.04	0.21 $\pm$ 0.16	0.53 $\pm$ 0.02	0.66 $\pm$ 0.01	0.63 $\pm$ 0.03	0.61 $\pm$ 0.03	0.31 $\pm$ 0.10	0.70 $\pm$ 0.00	0.57 $\pm$ 0.06
LLAMAFEWSHOT	0.39 $\pm$ 0.05	0.21 $\pm$ 0.04	0.43 $\pm$ 0.05	0.43 $\pm$ 0.01	0.60 $\pm$ 0.04	0.54 $\pm$ 0.07	0.38 $\pm$ 0.16	0.62 $\pm$ 0.04	0.36 $\pm$ 0.05
LLAMARAG	0.43 $\pm$ 0.01	0.16 $\pm$ 0.01	0.47 $\pm$ 0.01	0.42 $\pm$ 0.05	0.28 $\pm$ 0.02	0.51 $\pm$ 0.08	0.09 $\pm$ 0.04	0.62 $\pm$ 0.00	0.55 $\pm$ 0.04
LLAMARAGCoT	0.50 $\pm$ 0.10	0.27 $\pm$ 0.06	0.58 $\pm$ 0.05	0.63 $\pm$ 0.04	0.49 $\pm$ 0.04	0.61 $\pm$ 0.05	0.10 $\pm$ 0.19	0.68 $\pm$ 0.06	0.59 $\pm$ 0.05

(MCC). In particular, LLMs with multimodal feature representation, especially when paired with RAG+CoT prompting, achieve higher and more stable correlations with ground-truth scores across most surveys, indicating that their advantages extend beyond discrete category prediction to modeling underlying well-being gradients. LLM variants that incorporate facial and acoustic cues (e.g., LLAMATF0, LLAMATA0) achieve the highest Pearson correlations for distress-related constructs such as GAD-7 and SQS, reinforcing the importance of nonverbal signals for detecting distress that may not be explicitly verbalized. In contrast, regression models and VLMs exhibit weaker or more inconsistent correlations across surveys. Taken together, these findings show that the advantages of LLMs with multimodal features, particularly using retrieval-augmented reasoning, persist across complementary evaluation metrics, strengthening confidence in their robustness for well-being assessment beyond any single performance measure.

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